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Degree: B.Tech.

Roll no.:
Semester: 3rd

Course work
END-SEMESTER (SUMMER) EXAMINATION, 2026

Course code: CMMTC303

Course title: Applied Linear Algebra

Time: 3 Hours

Maximum Marks: 50

Note: Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed and mentioned in the answer. Here, we follow the same notations used in class.

Q. No.	Question	Marks	CO
Q. 1	Attempt any two parts of the following:		
1a	Consider the vector space $P_2(t)$ of polynomials of degree ≤ 2 . Show that the set $\{t+1, t-1, (t-1)^2\}$ forms a basis of $P_2(t)$. Also find a, b, c such that $2t^2 - 5t + 9 = a(t+1) + b(t-1) + c(t-1)^2.$	5	CO1
1b	Prove or disprove the vectors $u = (1+i, 2i)$ and $w = (1, 1+i)$ in $\mathbb{C}^2$ are linearly dependent over the complex field $\mathbb{C}$ but linearly independent over the real field $\mathbb{R}$.	5	CO1
1c	Suppose that $A_1, A_2, \dots$ are linearly independent sets of vectors, and that $A_1 \subseteq A_2 \subseteq \dots$. Show that the union $A = A_1 \cup A_2 \cup \dots$ is also linearly independent.	5	CO1
Q. 2	Attempt any two parts of the following:		
2a	Let T be the linear operator on $\mathbb{R}^2$ defined by $T(x, y) = (2x + y, 3x + 2y)$. Show that T is invertible and find its inverse.	5	CO2
2b	Let V be a vector space of all $n \times n$ matrices over field F , and B be a fixed $n \times n$ matrix. Define a map from V to V by $T(A) = AB - BA$ for all $A \in V$. Prove or disprove T is a linear map.	5	CO2
2c	Let V be the vector space of 2×2 matrices. Consider the following matrix M and usual basis E of V : $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ and } E = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$ Find the matrix representing of the linear operator $T(A) = MA - AM$ on V relative to E .	5	CO2
Q. 3	Attempt any two parts of the following:		
3a	Let A be an $n \times n$ matrix. Prove that $\dim(\text{span}(\{I_n, A, A^2, \dots\})) \leq n.$	5	CO3
3b	Let T be a non-zero linear operator on a vector space V . Then prove that a nilpotent operator is never diagonalizable.	5	CO3

Q. No.	Question	Marks	CO
3c	Let $A = \begin{pmatrix} 7 & 3 \\ 3 & -1 \end{pmatrix}$. Find an orthogonal matrix P such that $D = P^{-1}AP$ is diagonal.	5	CO3
Q. 4	Attempt any two parts of the following:		
4a	Let A be a square matrix with the property that the sum of elements in each of its row is 1. Show that $\lambda = 1$ is an eigenvalue of A , and also find an eigenvector corresponding to $\lambda = 1$.	5	CO4
4b	Verify the following polarisation identity in a real inner product space V : $\langle x, y \rangle = \frac{1}{4}(\ x+y\ ^2 - \ x-y\ ^2), \quad \forall x, y \in V.$	5	CO4
4c	Consider the vector space $P(t)$ with inner product $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$. Apply the Gram-Schmidt algorithm to the set $\{1, t, t^2\}$ to obtain an orthogonal set $\{f_0, f_1, f_2\}$ with integer coefficients.	5	CO4
Q. 5	Attempt any two parts of the following:		
5a	Define bilinear form. Let g be a bilinear form on $\mathbb{R}^n$. Let B be an ordered basis of $\mathbb{R}^n$, then show that g is given by $g(\alpha, \beta) = [\alpha]_B^t A [\beta]_B$ for some $n \times n$ matrix A , where $[\alpha]_B^t$ means transpose of $[\alpha]_B$.	5	CO5
5b	Let $M_n(\mathbb{C})$ be the space of $n \times n$ matrices over $\mathbb{C}$. Define a map $f : M_n(\mathbb{C}) \times M_n(\mathbb{C}) \rightarrow \mathbb{C}$ by $f(A, B) = n \text{tr}(AB) - \text{tr}(A) \text{tr}(B),$ where $\text{tr}(A)$ means trace of the matrix A . Prove or disprove f is a symmetric bilinear form.	5	CO5
5c	Define quadratic form. Let f be a symmetric bilinear form on a finite dimensional vector space V over F and q be the associated quadratic form. Then, show that $f(\alpha, \beta) = \frac{1}{4}(q(\alpha + \beta) - q(\alpha - \beta)) \quad \alpha, \beta \in V.$	5	CO5